Unlocking Behavioral Intelligence in Airline Loyalty Programs

Background

Airlines use loyalty programs to increase repeat business and improve Customer Lifetime Value (CLV).However, many programs still focus mainly on points and rewards. As a result, many members stay inactive, high-value customers may stop flying without being noticed, and marketing teams often react too late. Airlines now need a more proactive and data-driven loyalty strategy.

The Challenge

You are part of the Business Analytics team of a mid-sized airline. You will work with data for around 16,700 Canadian loyalty members from 2012 to 2018.
Your goal is to build a solution that helps the marketing team identify customers likely to disengage, understand which members are most valuable, and recommend actions to improve retention. The main The challenge is not only building a model, but also turning insights into clear business actions.

Project Roadmap

Phase 1: Data Architecture & Preparation

- **Data Ingestion:** Merged multi-year operational datasets covering Customer Loyalty History and Customer Flight Activity.
- **Defensive Cleaning:** Handled missing demographic values (e.g., median-based salary imputation) to prevent algorithmic bias.
- **Deduplication & Aggregation:** Cleaned data pipeline artifacts by resolving split-booking records and aggregating additive metrics to ensure exactly one row per customer-month.
- **Time-Series Standardization:** Converted raw month/year columns into cohesive datetime indexes to enable accurate recency and lifecycle calculations.

Phase 2: Churn Factors & Feature Engineering

- **Defining the "Churn" Anchor:** Replaced the flawed "cancellation-only" metric with a multi-factor behavioral definition of churn (e.g., flagging members with zero flights and redemptions over a 12-month period).
- **Behavioral Aggregation:** Engineered customer-level features to track engagement over time, including total lifetime flights, points accumulated, points redeemed, and months since last activity.
- **Derived Efficiency Metrics:** Created strategic velocity metrics such as "Points per KM flown" and "Redemption Rate" to measure depth of engagement.

Phase 3: Predictive Modeling (Churn Propensity)

- **Algorithm Deployment:** Trained a Random Forest Classifier to predict a user's future susceptibility to churn.

- **Leakage Prevention:** Purged the training dataset of any features that inherently defined churn (e.g., months inactive, cancellation dates) to ensure the model predicted true susceptibility rather than memorizing rules.
- **Class Imbalance Correction:** Applied class weighting to force the algorithm to accurately identify the minority class (churned users) without sacrificing overall precision.
- **Driver Extraction:** Analyzed model feature importances to identify the top behavioral and demographic drivers causing customer attrition.

Phase 4: Multidimensional Customer Segmentation

- **Beyond Basic CLV:** Recognized that historical Customer Lifetime Value (CLV) is a lagging indicator and combined it with forward-looking behavioral metrics (frequency, engagement intensity).
- **Unsupervised Learning:** Deployed a K-Means clustering algorithm to mathematically group the customer base into distinct cohorts based on their holistic flight and spend profiles.
- **Persona Profiling:** Translated the mathematical clusters into humanized, actionable business personas (e.g., VIP Loyalists, Emerging Members, At-Risk Dormant).

Phase 5: Strategic Execution & GTM Playbook

- **Risk Stratification:** Crossed the Churn Probability scores (from Phase 3) with the Customer Personas (from Phase 4) to identify high-value users at high risk of leaving.
- **Product-Led Interventions:** Developed specific, data-backed retention playbooks for each persona, shifting away from generic discount codes to structural product interventions (e.g., automated status matches, micro-redemptions).
- **Operational Dashboarding:** Designed a functional interface for marketing managers to effortlessly filter the data, identify at-risk cohorts, and instantly deploy the recommended retention campaigns.

Phase 1: Data Architecture & Defensive Cleaning

1. Demographic Sanitization (Customer Loyalty History)

Before modeling, demographic data was cleaned to remove nulls and standardize formats without distorting customer distribution.

- **Salary Imputation via Education Tiers**
  Missing Salary values were replaced using the **median salary within each Education group**, with a global median fallback if an entire tier was empty.
  This avoided skew from high-income outliers (e.g., executives) and prevented unrealistic assignments like CEO-level salaries to students. Rows were not dropped to avoid survivorship bias.
- **Standardizing Cancellation Flags & Strings**
  Cancellation Year and Month were converted from float to integer, replacing NaN values (active users) with 0. An `Is_Active` boolean flag was created, and all string columns (Province, City, etc.) were stripped and title-cased.

This ensured strict binary logic for ML models and prevented duplicate geographic categories caused by inconsistent formatting (e.g., `" toronto "` vs `"Toronto"`).

2. Behavioral Time-Series Consolidation (Customer Flight Activity)

The fragmented flight activity logs were transformed into a clean behavioral timeline.

- **Removing Pipeline Artifacts & Split Records**
  Over 1,900 exact duplicate rows were removed. Another ~1,900 rows with the same Customer-Year-Month but different metrics were aggregated by summing additive columns (Flights, Distance, Points).
  These represented booking amendments (e.g., upgrades after booking). Without consolidation, the model would incorrectly assume multiple flights in the same month. Aggregation enforced a strict "one row per customer-month" structure.
- **Defensive Clipping Against System Errors**
  All additive columns were clipped at a minimum of 0 to eliminate negative values.
  This prevented impossible cases like negative flights or distances caused by refunds or pipeline glitches, which could break production ML systems.
- **Temporal Formatting & Derived Efficiency**
  Year and Month were merged into proper Pandas datetime periods, and a `Points_Per_KM` feature was engineered with divide-by-zero handling.
  Datetime formatting enabled recency calculations for churn modeling, while `Points_Per_KM` helped identify reward-system gamers versus standard budget travelers—critical for later K-Means segmentation.

Phase 2: Churn Factors and Feature Engineering

1. Behavioral Aggregation (Building the Timeline)

Before we could define who was churning, we needed to establish exactly when each user last interacted with the airline.

- **Extracting Recency and Lifetime Totals**
  - **The What:** We grouped the activity dataset by `Loyalty Number` to extract the absolute maximum (most recent) date a user either booked a flight or redeemed points. We also summed their lifetime flights and redemptions. Finally, we calculated `Months_Inactive` by comparing their `Last_Active_Date` against a pinned `CURRENT_DATE`.
  - **The Why:** Recency is the single strongest indicator of retention. A user who hasn't flown in 11 months behaves very differently from one who flew last week. By pinpointing the exact `Months_Inactive`, we built the temporal baseline required to execute our "Silent Churn" rule.

2. Deriving the "Exception" Flags (Handling Edge Cases)

Real-world operational data is messy. Users might cancel a specific co-branded credit card, but keep flying with the airline using corporate travel portals.

- **The Post-Cancellation Flight Flag**
    - **The What:** After merging our behavioral features back into the master dataset, we engineered explicit boolean flags: `Is_Cancelled` and, most importantly, `Flew_After_Cancel`. This checks if a user's `Last_Flight_Date` occurred chronologically after their `Cancellation_Date`.
    - **The Why:** This protects the business from "False Positives." If we blindly trusted the administrative cancellation date, we would incorrectly label active revenue-generating flyers as "Churned." Engineering this flag allows the algorithm to recognize the nuance of user behavior over strict administrative statuses.

3. The Multi-Factor Churn Logic (The Core Engine)

This is the intellectual property of the project. We replaced the airline's standard, reactive cancellation metric with a proactive, behavioral definition of churn.

- **Cascading Rule Implementation**
    - **The What:** We defaulted every user in the database to **Active (0)**. We then applied a cascading series of boolean masks:
        - **Rule 1 (Silent Churn):** Overrides to **1** if the user has been inactive for 12+ months.
        - **Rule 2 (Hard Exit):** Overrides to **1** if they formally canceled and did not fly afterward.
        - **Rule 3 (The Exception):** Overrides back to **0** if they canceled but kept flying.
        - **Audit Trail:** We generated a `Churn_Reason` text column documenting exactly which rule triggered the final label.
    - **The Why:** This logic captures the true health of the loyalty program. It flags the "Silent Majority"—users who simply stop flying and forget about the airline who make up the bulk of the **16.26% churn rate**. Furthermore, adding the `Churn_Reason` audit trail ensures that the data engineering or marketing teams can seamlessly debug the logic and trust the labels before feeding them into a machine learning model.

Model Output :

```
Churn_Flag
Active (0)      83.74
Churned (1)     16.26
Name: proportion, dtype: float64
```

Phase 3: Predictive Modeling (Churn Propensity)

1. Leakage-Free Feature Engineering

Before feeding data into the machine learning model, we had to finalize the features, ensuring we didn't accidentally give the model the "answers" to the test.

- **Calculating Lifetime Metrics & Tenures**
    - **The What:** We aggregated `activity_df` to calculate lifetime totals (e.g., `Total_Distance_Ever`, `Months_Flown`). We also calculated `Tenure_Months` by comparing the `Enrollment_Date` to the pinned `CURRENT_DATE`.
    - **The Why:** We needed features that measure the *depth* and *efficiency* of a customer's engagement, not just raw totals. By dropping features that directly define the churn logic (like `Months_Inactive` or `Cancellation_Date`), we strictly prevented "Data Leakage." If we included those, the model would simply memorize the rules rather than learning underlying behavioral patterns.

2. Preprocessing & The Random Forest Architecture

Machine learning algorithms require clean, structured math. We had to transform categorical text and scale numerical values before training.

- **The Scikit-Learn Pipeline**
    - **The What:** We built a `ColumnTransformer` using an `OrdinalEncoder` for categorical variables (like `Province` or `Education`) and a `StandardScaler` for numeric variables. We then split the data into a 70/30 Train/Test set using `stratify=y` to maintain the 84/16 class imbalance.
    - **The Why:** Using an Ordinal Encoder instead of One-Hot Encoding prevents a massive expansion of columns, keeping the Random Forest model lightweight and fast. The `stratify` command is crucial; without it, the test set might randomly contain zero churned users, ruining our evaluation metrics.
- **Initializing the Balanced Random Forest**
    - **The What:** We initialized a `RandomForestClassifier` with shallow parameters (`max_depth=8`, `n_estimators=200`) and the critical argument: `class_weight="balanced"`.
    - **The Why:** Because only ~16% of users actually churned, a standard model would just guess "Active" every time and achieve 84% accuracy while failing its primary business goal. `class_weight="balanced"` mathematically forces the algorithm to pay 5x more attention to the minority (Churned) class, prioritizing the identification of at-risk users over raw accuracy.

3. Evaluation Metrics: Interpreting the Output

The model's performance on the unseen 30% test set proves its viability for real-world deployment.

- **ROC-AUC Score: 0.966**
    - A score of **0.966** is exceptionally high, indicating that the model has learned robust, distinct patterns differentiating active flyers from those about to churn. It proves to stakeholders that the model is highly trustworthy.
- **The Classification Report (Focusing on Class 1 - Churned)**
    - **Precision (0.83):** When the model flags a user as "Churned," it is correct 83% of the time. This means the marketing team won't waste significant budget sending retention offers to users who were never going to leave.

- **Recall (0.97):** Out of all the users who *actually* churned, the model successfully caught 97% of them. This is the most important metric for a retention strategy; the model is highly aggressive at finding at-risk users, minimizing missed opportunities.

4. Feature Importances: The Business "Why"

Machine learning is useless if it's a black box. We extracted the internal logic of the Random Forest to understand what actually drives churn.

- **The Top 5 Drivers**
  1. `num__Months_Flown` (18.88%)
  2. `num__Total_Flights_Ever` (16.59%)
  3. `num__Tenure_Months` (11.13%)
  4. `num__Redemption_Rate` (9.41%)
  5. `num__Avg_Flights_Per_Month` (8.85%)
- **The Strategic Justification:** Notice that **none** of the demographic features (Salary, Education, Province) made the top five. The algorithm proves definitively that churn is a **behavioral habit problem, not a demographic one.** Users churn because they fail to build a consistent cadence of flying (`Months_Flown`) and fail to experience the reward loop (`Redemption_Rate`), completely validating our strategic pivot away from generic discounts toward Product-Led interventions.

5. Risk Stratification

- **The What:** We used `model.predict_proba()` to assign a continuous churn probability score (0.0 to 1.0) to every user in the master database. We then bucketed them into "Low," "Medium," and "High" risk segments.
- **The Why:** Marketing managers cannot execute campaigns based on raw probabilities like "0.7432." Bucketing users into defined risk segments provides a clean, operational trigger for the dashboard, allowing the team to immediately filter and target the "High Risk" cohort.

Model Output:

```
ROC-AUC Score: 0.965

--- Classification Report ---
              precision    recall  f1-score   support

           0       0.98      0.96      0.97      4206
           1       0.83      0.88      0.85       816

    accuracy                           0.95      5022
   macro avg       0.90      0.92      0.91      5022
weighted avg       0.95      0.95      0.95      5022


--- Top 5 Drivers of Churn ---
                  Feature  Importance
        num__Months_Flown    0.259776
  num__Total_Flights_Ever    0.173839
num__Avg_Flights_Per_Month   0.138564
 num__Total_Distance_Ever    0.124920
       num__Tenure_Months    0.111191

── Risk segment distribution ───────────────────────────────
              Count  Actual_Churners  Churn_Rate_pct
Risk_Segment
Low Risk      12784              177             1.4
Medium Risk    1319              208            15.8
High Risk      2634             2336            88.7

✓ Saved → master_churn_scored.csv
```

Phase 4: Multidimensional Customer Segmentation

1. Engineering Velocity & Intensity Metrics

Before grouping our customers, we had to address the "CLV Paradox"—the realization that raw lifetime totals (like historical CLV or Total Flights) heavily bias the algorithm toward older members while ignoring high-potential new members.

- **Calculating Penetration and Intensity**
  - **The What:** We engineered new columns to measure *rates* rather than *totals*. `CLV_Per_Month` tracks value generation speed. `Frequency_Penetration` tracks the percentage of active months out of the 24-month observation window. Most importantly, `Engagement_Intensity` tracks how many flights a user takes *only during the months they actually fly* (avoiding division by zero).

- **The Why:** These velocity metrics normalize the data. A corporate traveler who joined 3 months ago but flies 4 times a month has massive "Engagement Intensity." If we only looked at "Total Flights," they would look identical to a budget flyer who took 12 flights over 5 years. This engineering step is what allows us to discover the "Future VIPs."

2. Standardization (Solving the K-Means Distance Problem)

Machine learning algorithms like K-Means are highly sensitive to the scale of the data.

- **The StandardScaler Implementation**
  - **The What:** We isolated our **10 chosen dimensions** (blending demographics, behavior, and our predicted `Churn_Probability`). We then passed them through Scikit-Learn's `StandardScaler` to transform all values so they have a mean of 0 and a standard deviation of 1.
  - **The Why:** K-Means groups customers by calculating the "Euclidean distance" between data points. If we didn't scale the data, `Salary` (measured in the tens of thousands) would completely overpower `Redemption_Rate` (measured in decimals like 0.15). Scaling ensures every feature has an equal mathematical vote in shaping the personas.

4. Translating Math into Personas (The Business Output)

The algorithm outputs abstract numbers (Cluster 0, 1, 2...). We had to translate these mathematical coordinates back into a boardroom-ready product strategy.

- **Persona Mapping & Profiling**
  - **The What:** We mapped the raw `Cluster_ID` to humanized names
    1. **VIP Loyalists:** The highest value, high-tenure corporate travelers who fly consistently and present a near-zero churn risk.
    2. Emerging Members: Newer members with moderate value who are still establishing their travel habits and are highly vulnerable to early-stage churn.
    3. Loyal Budget Flyers: The massive volume engine of the business; long-tenure customers with lower individual CLV but highly consistent flight patterns.
    4. At-Risk Dormant: Previously high-value members who have not flown in months and require aggressive win-back campaigns.
    5. High-Engagement Newcomers: The future VIP pipeline; brand new members showing extremely high flight intensity right out of the gate.
    using a dictionary. We then grouped the entire dataset by these new personas to calculate the average stats for each cohort, verifying the results against our initial hypothesis.
  - **The Why:** A CMO cannot launch a marketing campaign targeting "Cluster 2." But they *can* launch a campaign targeting "Loyal Budget Flyers." By appending the `Customer_Persona` directly onto the master dataset, we created the exact filtering mechanism needed for our final Streamlit operational dashboard.

Model Output :

```
--- FINAL CLUSTER PROFILES ---
Customer_Persona       At-Risk Dormant  Emerging Members  High-Engagement Newcomers  Loyal Budget Flyers  VIP Loyalists
CLV                            8788.49           6205.26                    7497.52              8056.82        9106.46
CLV_Per_Month                  1393.08            169.35                     365.76               275.91         242.10
Tenure_Months                     8.19             45.27                      39.12                39.55          46.06
Frequency_Penetration             0.21              0.58                       0.47                 0.04           0.59
Engagement_Intensity              4.33              2.67                       2.90                 0.70           2.69
Redemption_Rate                   0.02              0.02                       0.02                 0.01           0.02
Recency_Months                    1.12              0.76                       2.62                21.94           0.74
Salary                        71302.01          74369.89                  226791.34             74897.72       74443.15
Card_Tier_Ordinal                 1.77              1.00                       1.71                 1.76           2.37
Churn_Probability                 0.34              0.09                       0.22                 0.97           0.08
Count                          3119.00           5107.00                     427.00              2105.00        5979.00
% of Base                        18.60             30.50                       2.60                12.60          35.70
```

Phase 5: Strategic Execution & GTM Playbook

0. VIP Loyalists (The Core Anchor)

The structural challenge here is not retention, but leveraging their loyalty to drive organic acquisition and increase switching costs.

- **Strategy 1: Status Portability (The Viral Loop)**
  - **The Action:** Allow VIPs to "gift" a 72-hour Nova pass (lounge access, priority boarding) to a colleague or family member twice a year.
  - **The Rationale:** It costs the airline almost nothing in hard dollars, but it makes the VIP feel immensely valued while serving as a high-conversion, zero-CAC (Customer Acquisition Cost) channel to acquire new business travelers.
- **Strategy 2: The "Zero-Friction" Product Guarantee**
  - **The Action:** Build a dedicated UI flow for VIPs. If a flight is delayed by more than 45 minutes, trigger an automated push notification allowing them to instantly rebook on the next flight with one tap—before they even reach the customer service desk.
  - **The Rationale:** VIPs value time above all else. Removing operational friction creates an emotional lock-in that competitors cannot break with simple point-matching.

1. Emerging Members (The Vulnerable Pipeline)

The challenge is bridging the "Time-to-Value" gap. If they do not experience a reward quickly, they will churn.

- **Strategy 1: "Phantom Points" & Loss Aversion**
  - **The Action:** Instead of a standard welcome email, drop 5,000 "Phantom Points" into their account upon sign-up. Add a highly visible countdown timer in the app showing these points will expire in 60 days unless they book their first flight.
  - **The Rationale:** Behavioral economics proves that loss aversion (fear of losing something you already have) is twice as powerful as the desire to gain something new.
- **Strategy 2: Micro-Milestone UI**
  - **The Action:** Redesign their app dashboard to show a progress bar not just for the next Tier, but for micro-rewards. (e.g., "You are 1 flight away from a free inflight Wi-Fi pass!").

    - **The Rationale:** Waiting a year to hit Nova tier is too abstract for a new user. Micro-milestones create an immediate dopamine loop.

2. Loyal Budget Flyers (The Volume Engine)

The challenge is margin expansion. We need to increase their monetary value without raising ticket prices, which would cause them to churn.

- **Strategy 1: In-Flight Micro-Burns (Liability Clearance)**
    - **The Action:** Introduce a feature allowing users to burn small amounts of points (e.g., 500 points) seamlessly via the app for premium snacks, headphones, or seat upgrades at the gate.
    - **The Rationale:** This segment currently has a near-zero redemption rate. These micro-burns clear point liabilities off the CFO's balance sheet while making the user feel like they are getting consistent value.
- **Strategy 2: The "Flight Pass" Subscription Model**
    - **The Action:** Offer a $9/month subscription tier that grants waived baggage fees, double points on off-peak flights, and exclusive 24-hour early access to seat sales.
    - **The Rationale:** Transforms episodic, unpredictable budget flyers into a source of guaranteed Monthly Recurring Revenue (MRR), while ensuring they never check a competitor's website first.

3. At-Risk Dormant (The Recovery Targets)

Standard win-back discounts rarely work long-term. We need algorithmic, highly contextual triggers.

- **Strategy 1: Algorithmic Route-Specific Price Alerts**
    - **The Action:** Do not send generic 20% off coupons. If the data shows a dormant user historically flew YYZ -> JFK every November, configure the system to send an automated push notification in October *only* if the price for that specific route drops below a certain threshold.
    - **The Rationale:** It feels personalized and highly relevant, cutting through standard promotional noise.
- **Strategy 2: The "Prove It" Status Challenge**
    - **The Action:** For high-CLV dormant users who likely left for a competitor, offer an instant 90-day status match to their current tier. To keep it for the rest of the year, they must complete 2 round trips in that 90-day window.
    - **The Rationale:** It lowers the barrier to re-entry while ensuring the airline gets guaranteed revenue to justify the perk.

4. High-Engagement Newcomers (The Future VIPs)

The challenge is cementing their habit before their initial surge in travel ends.

- **Strategy 1: The Fast-Track Multiplier**
    - **The Action:** Detect when a new user completes 3 flights in their first 45 days. Instantly trigger a "Fast-Track" UI takeover in the app, offering them 3x points on all flights for the next 90 days.

    - **The Rationale:** This exploits their current high-velocity momentum, forcing them to consolidate all their upcoming travel onto your airline to maximize the multiplier.
- **Strategy 2: "Business Mode" App Toggle**
    - **The Action:** Recognizing these are likely new corporate travelers, introduce a one-tap "Business Mode" in the app that auto-generates expense reports, separates personal/business points, and integrates with Concur/Expensify.
    - **The Rationale:** By solving a massive pain point for business travelers (expensing), you embed the airline into their professional workflow, making it incredibly annoying for them to switch to a competitor.